Polylogarithmic Polynomial Commitment Scheme over Galois Rings

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Abstract. This paper introduces the first multilinear polynomial commitment scheme (PCS) over Galois rings achieving $\mathcal{O}(\log^2 n)$ verification cost. It achieves $\mathcal{O}(n \log n)$ committing time and $\mathcal{O}(n)$ evaluation opening prover time. This PCS can be used to construct zero-knowledge proofs for arithmetic circuits over Galois rings, facilitating verifiable computation in applications requiring proofs of polynomial ring operations (e.g., verifiable fully homomorphic encryption). First we construct random foldable linear codes over Galois rings with sufficient code distance and present a distance preservation theorem over Galois rings. Second we extend the Basefold commitment (Zeilberger et al., Crypto 2024) to multilinear polynomials over Galois rings. Our approach reduces proof size and verifier time from $\mathcal{O}(\sqrt{n})$ to $\mathcal{O}(\log^2 n)$ compared to Wei et al., PKC 2025. Furthermore, we give a batched multipoint openning protocol for evaluation phase that collapses the proof size and verifier time of N polynomials at M points from $\mathcal{O}(NM \log^2 n)$ to $\mathcal{O}(\log^2 n)$, prover time from $\mathcal{O}(NMn)$ to $\mathcal{O}(n)$, further enhancing efficiency.

Keywords: Polynomial Commitment · Galois Ring · zk-SNARK.

1 Introduction

Polynomial commitment schemes serve as foundational building blocks for zero-knowledge succinct non-interactive argument of knowledge (zk-SNARKs)(e.g. [1,6,8,11,18]), with critical applications in privacy-preserving computation and blockchain systems. Existing PCS constructions primarily operate over finite fields. However, the emergence of lattice-based cryptographic protocols involving polynomial rings necessitates efficient PCS constructions over Galois rings. These non-field rings introduce unique challenges, including zero divisors and complex distance metrics, which render conventional finite-field-based techniques inapplicable. Despite recent progress in building proofs over polynomial rings[10,17,12,7], they extended Brakedown PCS [11] and Pinocchio [14] to rings, no prior works achieve logarithmic verification cost for ring arithmetic circuits without trusted setup. While Rinocchio [10] offers constant verification, it necessitates a circuit-specific trusted setup, and each proof for different circuit requires executing a precompile process. Our PCS is transparent, requiring only a single initial setup that is applicable to commitments over different polynomials.

Motivation. Existing PCS over Galois rings face two primary limitations: (1) verification complexity are limited to $\mathcal{O}(\sqrt{n})$, and (2) they lack support for batched evaluation of multiple polynomials and points. To achieve a more efficient PCS, we extend the Basefold commitment scheme for polynomials over finite fields [18] to polynomials over Galois rings. And we present a bached variant for Basefold to support multi-point opening.

Our Approach. Our goal is to construct a scheme for committing to a multilinear polynomial f over Galois rings and proving the correctness of its evaluation at a point x, i.e., $f(x) = y$. A multilinear polynomial is a multivariate polynomial that the degree of the polynomial in each variable is at most one. Any multilinear polynomial $f(X_1, X_2, \cdots, X_d)$ can be uniquely expressed as $\sum_{\mathbf{b}\in\{0,1\}^d} f(\mathbf{b})\tilde{eq}_{\mathbf{b}}(X_1, X_2, \cdots, X_d)$, where $\tilde{eq}_{\mathbf{b}}$ is the multilinear Lagrange basis polynomial with interpolating set $\{0,1\}^d$. Thus, proving $f(x) = y$ reduces to proving a sum. We leverage the sum-check protocol ([13], Chapter 4 from [16]) to prove this sum. However, the final step of the sum-check protocol requires evaluating f at a random point. Here, we introduce IOPP[15,2]. In this context, an IOPP is an interactive proof (in random oracle model) demonstrating that a vector is close to a codeword in a given code. A key observation in the Basefold commitment construction[18], is that IOPP execution can be viewed as opening a polynomial at a random point. This directly addresses the need to evaluate f at a random point in the final step of the sum-check protocol. We exploit this insight to construct a PCS for multilinear polynomials over Galois rings.

Our Contributions. The main contributions of this work are as follows:

- **Random Foldable Codes over Galois Rings:** By constructing (c, k_0, d)-foldable linear codes with provable minimum distance bounds via homomorphic reduction to finite fields (Lemma 1), we enable $\mathcal{O}(n \log n)$-time encoding and sufficient minimum code distance for following constructions.
- **Basefold Polynomial Commitment over Galois rings:** By introducing a linear proximity gap theorem (Theorem 1) for linear codes over Galois rings and combining IOPP with the sum-check protocol, we obtain a multilinear PCS(Figure 3) over Galois rings with $\mathcal{O}(\log^2 n)$ proof size and verifier time, $\mathcal{O}(n \log n)$ committing time, and $\mathcal{O}(n)$ evaluation opening proof time.
- **Multipoint opening Protocol:** A batched evaluation protocol leveraging polynomial linearization (Figure 4) to reduce N-polynomial M-point verification costs from $\mathcal{O}(NM \log^2 n)$ to $\mathcal{O}(\log^2 n + N + M)$, and prover cost from $\mathcal{O}(NMn)$ to $\mathcal{O}(n + N + M)$, thereby further enhancing efficiency.

2 Preliminaries

Definition 1 (Galois Ring). *Let p be a prime number and $s, r \in \mathbb{N}$. The Galois ring $GR = GR(p^s, r)$ is the finite commutative ring with p^{sr} elements, constructed as: $GR(p^s, r) \cong \mathbb{Z}_{p^s}[X]/(f(X))$, where $f(X)$ is a monic polynomial of degree r which is irreducible modulo p. We denote $GR^\times(p^s, r)$ as the set of all invertible elements in $GR(p^s, r)$, and we have $|GR^\times(p^s, r)| = p^{r(s-1)}(p^r - 1)$.*

Definition 2 (Multilinear Polynomial over Galois Rings). *A multilinear polynomial f in d variables $X_1, \ldots, X_d$ over Galois ring $GR(p^s, r)$ is represented as $f \in GR(p^s, r)[X_1, \ldots, X_d]$, where the coefficient vector $\mathbf{f} \in GR(p^s, r)^{2^d}$ encodes the 2^d coefficients, corresponding to all possible terms with each variable X_i appearing to degree 0 or 1.*

Definition 3 (Relative Minimum Distance of Linear Codes over Galois Rings). *Let C be an $[n, k, d]$-linear code over the Galois ring $GR(p^s, r)$, where n is the codeword length, k is the message length and d is the minimum Hamming distance. The Hamming distance between two codewords $\mathbf{c}, \mathbf{d} \in C$, denoted by $\Delta(\mathbf{c}, \mathbf{d})$, is the number of positions where they differ: $\Delta(\mathbf{c}, \mathbf{d}) = |\{i \mid c_i \neq d_i, 1 \leq i \leq N\}|$. The relative minimum distance Δ_C of code C is:*

$$\Delta_C = \frac{\min\{\Delta(\mathbf{c}, \mathbf{d}) \mid \mathbf{c}, \mathbf{d} \in C, \mathbf{c} \neq \mathbf{d}\}}{N}.$$

Definition 4 (Coset Relative Minimum Distance [18]). *Let n be an even integer and let C be an $[n, k, d]$-linear code over $GR(p^s, r)$. Let $\mathbf{v} \in GR(p^s, r)^n$ be a vector and let $\mathbf{c} \in C$ be a codeword. The coset relative distance $\Delta^*(\mathbf{v}, \mathbf{c})$ between $\mathbf{v}$ and $\mathbf{c}$ is defined as:*

$$\Delta^*(\mathbf{v}, \mathbf{c}) = \frac{2\,|\{j \in [1, n/2] : \mathbf{v}[j] \neq \mathbf{c}[j] \vee \mathbf{v}[j + n/2] \neq \mathbf{c}[j + n/2]\}|}{n}.$$

The relative minimum distance of $\mathbf{v} \in GR(p^s, r)^n$ to the code C is defined as: $\Delta^*(\mathbf{v}, C) = \min_{\mathbf{c} \in C} \Delta^*(\mathbf{v}, \mathbf{c})$, and $\Delta(\mathbf{v}, C) \leq \Delta^*(\mathbf{v}, C)$ (lemma 6 [18]).

Definition 5 (Maximum Distance Separable Code). *A linear code with length n, dimension k, and minimum distance d satisfies $d \leq n - k + 1$ (the Singleton Bound). It is called* Maximum Distance Separable (MDS) *if and only if $d = n - k + 1$.*

Definition 6 (Johnson Bound [18]). *For every $\gamma \in (0, 1)$, define $J_\gamma : [0, 1] \to [0, 1]$ as the function $J_\gamma(\lambda) := 1 - \sqrt{1 - \lambda(1 - \gamma)}$.*

Definition 7 (Free Module). *Let R be a ring. An R-module F is called free if there exists a subset $B \subseteq F$, called a basis, such that:*

1. *B is linearly independent: for any distinct $b_1, \ldots, b_n \in B$ and $r_1, \ldots, r_n \in R$,*

$$r_1 b_1 + \cdots + r_n b_n = 0_F \implies r_1 = \cdots = r_n = 0_R.$$

2. *B spans F: every $x \in F$ can be expressed as*

$$x = s_1 b_1 + \cdots + s_m b_m,$$

for some $b_1, \ldots, b_m \in B$ and $s_1, \ldots, s_m \in R$.

Free modules are defined to capture and generalize the essential property of vector spaces over fields. A free module over a ring extends the notions of "basis" and "linear combination" to the more general setting of modules.

Definition 8 (Polynomial commitment scheme [11]). *A polynomial commitment scheme for multilinear polynomials over a Galois ring $GR(p^s, r)$ is a tuple of four protocols PC = (Gen, Commit, Open, Eval):*

- *Gen$(1^\lambda, \mu)$: Generate public parameters pp, taking security parameter λ and number of variables μ as input.*
- *Commit(pp, G, γ): Takes public parameters pp, randomness γ and a μ-variate multilinear polynomial $G \in GR(p^s, r)[\mu]$, outputs a commitment C.*
- *Open(pp, C, G, γ): Verifies if commitment C corresponds to polynomial G and randomness γ, outputs a boolean $b \in \{0, 1\}$.*
- *Eval(pp, C, z, v, μ, G): A protocol between a prover $\mathcal{P}$ and verifier $\mathcal{V}$. $\mathcal{P}$ and $\mathcal{V}$ hold $pp, C, z \in GR(p^s, r)^\mu, v \in GR(p^s, r), \mu$, only $\mathcal{P}$ holds G. $\mathcal{P}$ attempts to convince $\mathcal{V}$ that $G(z) = v$. The verifier outputs a boolean $b \in \{0, 1\}$.*

A tuple of four protocols (Gen, Commit, Open, Eval) is an extractable polynomial commitment scheme for multilinear polynomials over a Galois ring $GR(p^s, r)$ if the following conditions hold:

- ***Completeness.*** *For any valid polynomial G and evaluation point r, if $v = G(r)$, an honest prover can convince the verifier to output 1 in the Eval protocol with probability at least 1.*
- ***Binding.*** *It is computationally hard for any PPT adversary to find a commitment C and two distinct polynomials $G_0 \neq G_1$ such that Open$(pp, C, G_0) = 1$ and Open$(pp, C, G_1) = 1$, except with negligible probability ($\leq negl(\lambda)$).*
- ***Knowledge Soundness.*** *Eval is a succinct argument of knowledge for the NP relation $R_{Eval} \subseteq (\{0, 1\}^* \times GR(p^s, r)^\mu \times GR(p^s, r)) \times GR(p^s, r)[\mu]$ defined as: A pair $((C, r, v), G)$ is in R_{Eval} if and only if $G \in GR(p^s, r)[\mu]$, $G(r) = v$, and Open$(pp, C, G) = 1$. The statement is (C, r, v) and the witness is G.*

Definition 9 (Foldable linear codes over Galois rings adapted from definition 5 in [18]). *For Galois ring $GR(p^s, r)$ and let $c, k_0, d \in \mathbb{N}$. A linear code $C_d : GR(p^s, r)^{k_0 \cdot 2^d} \to GR(p^s, r)^{ck_0 \cdot 2^d}$ with generator matrix $\mathbf{G}_d$ over Galois ring $GR(p^s, r)$ is called* foldable *if there exists a list of matrices $(\mathbf{G}_0, \ldots, \mathbf{G}_{d-1})$ and diagonal matrices $(T_0, \ldots, T_{d-1})$ and $(T'_0, \ldots, T'_{d-1})$ over $GR(p^s, r)$, such that for every $i \in [1, d]$ satisfy 1,2 and for every $i \in [0, d]$ satisfy 3:*

1. *The diagonal matrices $T_{i-1}, T'_{i-1} \in (GR(p^s, r))^{ck_0 \cdot 2^{i-1} \times ck_0 \cdot 2^{i-1}}$ satisfies that diag$(T_{i-1})[j] \neq$ diag$(T'_{i-1})[j]$ for every $j \in [ck_0 \cdot 2^{i-1}]$;*
2. *The matrix $\mathbf{G}_i \in (GR(p^s, r))^{ck_0 \cdot 2^i \times ck_0 \cdot 2^i}$ equals:*

$$\mathbf{G}_i = \begin{bmatrix} \mathbf{G}_{i-1} & \mathbf{G}_{i-1} \\ \mathbf{G}_{i-1} \cdot T_{i-1} & \mathbf{G}_{i-1} \cdot T'_{i-1} \end{bmatrix}.$$

3. *$\mathsf{Im}(\mathbf{G}_i) = \{m \cdot \mathbf{G}_i \mid m \in GR(p^s, r)^{k_0 \cdot 2^i}\}$ are free GR-modules.*

Fact 1 *For Galois ring $GR(p^s, r)$ and let $a, b \in GR(p^s, r)$ with $a, b \neq 0$. Then the linear equation $ax = b$ has at most $p^{r(s-1)}$ solutions.*

Fact 2 *For Galois ring $GR(p^s, r)$ and let $f(x)$ be a non-zero polynomial of degree d over $GR(p^s, r)$. Then for a randomly chosen element $x \in GR(p^s, r)$, the probability that $f(x) = 0$ is at most $\frac{d}{p^r}$.*

3 Random Foldable Codes over Galois Rings

In this section, we present a foldable linear code over Galois rings. We present its encoding algorithm and analysis of its relative minimum distance.

Definition 10 (Random Foldable Linear Codes). *Fixed the Galois ring $GR(p^s, r)$, $c, k_0 \in \mathbb{N}$. For $i = 0$, define $\mathbf{G_0}$ as a one row, c columns matrix: $(v_1, v_2, \cdots, v_c)$, where $v_j \in GR^{\times}(p^s, r)$ for $1 \leq j \leq c$. For every $i > 0$, define foldable linear codes by sampling generator matrices $(\mathbf{G_1}, \mathbf{G_2}, \cdots, \mathbf{G_i})$ as follows:*

Sample $\mathsf{diag}(T_{i-1}) \xleftarrow{\$} GR^{\times}(p^s, r)$ and define $\mathbf{G_i}$ as

$$\mathbf{G_i} = \begin{bmatrix} \mathbf{G_{i-1}} & \mathbf{G_{i-1}} \\ \mathbf{G_{i-1}} \cdot T_{i-1} & \mathbf{G_{i-1}} \cdot (-T_{i-1}) \end{bmatrix}$$

where $\mathbf{G_i} \in (GR^{\times}(p^s, r))^{k_i \times n_{i-1}}$, $k_i = 2^i k_0$,$n_i = ck_i$ ($n_0 = c$, $k_0 = 1$). For a $d \in \mathbb{N}$, denote C_d as the $[c, k_0, d]$-foldable linear code with generator matrix $\mathbf{G_d}$.

For $0 \leq i \leq d$, since the entries of each G_i are invertible, we can perform Gaussian elimination on $\mathbf{G_i}$ to obtain its row-echelon form. The columns within the row-echelon submatrix of $\mathbf{G_i}$ then form a basis. The span of this basis in $\mathrm{GR}(p^s, r)$ is the set $\mathsf{Im}(\mathbf{G_i}) = \{m \cdot \mathbf{G_i} \mid m \in \mathrm{GR}(p^s, r)^{k_i}\}$, which constitutes a free GR-module (Definition 7).

Encoding algorithms for random foldable codes. The encoding logic of this algorithm is based on a divide-and-conquer recursion, achieving an encoding complexity of $\mathcal{O}(n \log n)$, where n denotes the codeword length. Specifically, to encode $m \in \mathrm{GR}(p^s, r)^{k_d}$ with generator matrix $\mathbf{G_d}$, the computation cost is $(1 + \frac{d}{2}) \cdot n_d$ multiplications and n_d additions over Galois ring $\mathrm{GR}(p^s, r)$.

Algorithm 1: Encoding of Random Foldable Codes over $\mathrm{GR}(p^s, r)$

Input:$\mathbf{m} \in \mathrm{GR}(p^s, r)^{k_d}$.
output:$\mathbf{w} \in \mathrm{GR}(p^s, r)^{n_d}$ such that $\mathbf{w} = \mathsf{Enc}_d(\mathbf{m}) = \mathbf{m} \cdot \mathbf{G}_d$.
Parameters: $\mathbf{G}_0$ and Diagonal matrices $(T_0, T_1, \ldots, T_{d-1})$ over $\mathrm{GR}(p^s, r)$.

if $d = 0$ **then**
  return $\mathsf{Enc}_0(\mathbf{m})$
else
  1. parse $\mathbf{m} = (\mathbf{m}_l, \mathbf{m}_r)$ where $\mathbf{m}_l, \mathbf{m}_r \in \mathrm{GR}(p^s, r)^{k_{d-1}}$ are of equal length;
  2. set: $\mathbf{l} \leftarrow \mathsf{Enc}_{d-1}(\mathbf{m}_l)$; $\mathbf{r} \leftarrow \mathsf{Enc}_{d-1}(\mathbf{m}_r)$; $\mathbf{t} \leftarrow \mathsf{diag}(T_{d-1})$;
  3. return $(\mathbf{l} + \mathbf{t} \circ \mathbf{r}, \mathbf{l} - \mathbf{t} \circ \mathbf{r})$.

3.1 Relative Minimum Distance

Directly computing the minimum code distance for large linear codes is computationally intractable. Thus, we theoretically analyze a lower bound on the

code distance. However, zero divisors significantly impact the probability of zero elements appearing in codewords. Precisely characterizing the distribution of zero divisors in codewords is laborious with limited gains, as different types of zero divisors belong to distinct ideals with varying proportions. Therefore, we take a different approach: demonstrating that the minimum distance of codes over $\mathrm{GR}(p^s, r)$ is no less than the minimum distance of foldable random linear codes over $\mathrm{GR}(p, r)$ which is a finite field of size $|p^r|$. In essence, this approach is equivalent to treating zero divisors within $\mathrm{GR}(p^s, r)$ as zero elements, which slightly reduces the analyzed minimum distance. The actual minimum distance will be somewhat larger, but this difference is minor, because in practice zero divisors constitute only a very small fraction of the elements in the ring.

For a linear code, analyzing the minimum distance between any two codewords is equivalent to evaluating the maximum number of zero entries in any individual codeword. Therefore, we present an analysis of the number of zero entries in random foldable linear codes over Galois rings.

Lemma 1. *For $v \in GR(p^s, r)^k$, let $v' = v \pmod p$. Let $\mathsf{num0}(v)$ denote the number of zero entries in v, i.e., $\mathsf{num0}(v) = |\{i \mid v_i = 0\}|$. Then, we have $\mathsf{num0}(v) \leq \mathsf{num0}(v')$.*

Proof. Let $v' = v \pmod p$. If $v_i = 0$ in $\mathrm{GR}(p^s, r)$, then $v'_i = v_i \pmod p = 0 \pmod p = 0$ in $\mathrm{GR}(p, r)$. Thus, the set of indices $\{i \mid v_i = 0\}$ is a subset of $\{i \mid v'_i = 0\}$. Taking the cardinality of these sets gives $\mathsf{num0}(v) \leq \mathsf{num0}(v')$. □

For (c, k_0, d)-foldable linear code C_d generated by $\mathbf{G_0}, T_0, T_1, \ldots, T_{d-1}$, given any $\mathbf{m} \in \mathrm{GR}(p^s, r)^{k_d} \setminus \mathbf{0}$, let $e = \mathsf{Enc}_d(\mathbf{m})$, and by definition 10 $e_p = e \mod p$ is:

$$\mathbf{m} \cdot \mathbf{G_d} \mod p = (\mathbf{m} \mod p) \cdot \begin{pmatrix} \mathbf{G_{d-1}} \mod p & \mathbf{G_{d-1}} \mod p \\ \mathbf{G_{d-1}} \cdot T_{d-1} \mod p & \mathbf{G_{d-1}} \cdot (-T_{d-1}) \mod p \end{pmatrix}.$$

By Lemma 1, we have $\mathsf{num0}(e) \leq \mathsf{num0}(e_p)$, and the relative minimum distance of Δ_{C_d} has a lower bound:

$$\Delta C_d = \frac{ck_d - \mathsf{num0}(e)}{ck_d} \geq \frac{ck_d - \mathsf{num0}(e_p)}{ck_d},$$

Next, to lower bound Δ_{C_d}, we derive the following upper bound on $\mathsf{num0}(e_p)$:

$$\mathsf{num0}(e_p) \leq ck_d \left(\frac{{\epsilon_{\mathrm{GR}}}^d}{c} + \frac{\epsilon_{\mathrm{GR}}}{\log p^r} \sum_{i=1}^{d} (\epsilon_{\mathrm{GR}})^{d-i} \left(0.6 + \frac{2\log(ck_i/2) + \lambda}{ck_i} \right) \right).$$

where $\epsilon_{\mathrm{GR}} = \frac{\log p^r}{\log p^r - 1.001}$. This bound follows directly from Theorem 2 in [18].

Let $\pi : \mathrm{GR}(p^s, r) \to \mathrm{GR}(p, r) \cong \mathbb{F}_{p^r}$ be the natural ring homomorphism defined by reduction modulo p. We have $\pi(ab) = \pi(a) \cdot \pi(b)$ for $a, b \in \mathrm{GR}(p^s, r)$. Specifically, if $\mathrm{GR}(p^s, r) \cong \mathbb{Z}_{p^s}[x]/(f(X))$ where $f(X)$ is a monic basic irreducible polynomial of degree r over $\mathbb{Z}_{p^s}[X]$, then π can be viewed as reducing the coefficients of polynomials in $\mathbb{Z}_{p^s}[X]/(f(X))$ modulo p to obtain elements

in $\mathbb{Z}_p[X]/(f(X)) \cong \mathrm{GR}(p,r) \cong \mathbb{F}_{p^r}$. $\mathrm{GR}(p,r)$ can also be treated as a finite field $\mathbb{F}_{p^r}$, which is the extension field of $\mathbb{F}_p$.

Let $\mathbf{m}' = \pi(\mathbf{m})$, we have $\mathbf{m}' \in \mathbb{F}_{p^r}^{k_d}$. Next, we consider linear codes C'_d over the finite field $\mathbb{F}_{p^r}$. The generator matrix of C'_d is $\mathbf{G'_d}$ constructed as follows:

Let $\mathbf{G'_0} = \pi(\mathbf{G_0})$ and codes generated by $\mathbf{G'_0}$ are MDS (Definition 5)[3]. Recursively generate $\{\mathbf{G'_i}\}_{1\le i\le d}$ as:

$$\mathbf{G'_i} = \begin{bmatrix} \mathbf{G'_{i-1}} & \mathbf{G'_{i-1}} \\ \mathbf{G'_{i-1}} \cdot \pi(T_{i-1}) & \mathbf{G'_{i-1}} \cdot (-\pi(T_{i-1})) \end{bmatrix} = \begin{bmatrix} \pi(\mathbf{G_{i-1}}) & \pi(\mathbf{G_{i-1}}) \\ \pi(\mathbf{G_{i-1}} \cdot T_{i-1}) & \pi(\mathbf{G_{i-1}} \cdot (-T_{i-1})) \end{bmatrix}$$

To show C'_d is (c, k_0, d)-foldable random linear code over $\mathbb{F}_{p^r}$ with random folding elements from $\mathbb{F}_{p^r}^{\times}$ (Definition 9 in [18]), we need to prove that if T_i is uniformly sampled from $\mathrm{GR}^{\times}(p^s, r)$, then $\pi(T_i)$ is uniformly distributed over $\mathbb{F}_{p^r}^{\times}$.

For any $\alpha \in \mathbb{F}_{p^r}^{\times}$, α can be written as a polynomial: $a_0 + a_1X + \cdots + a_{r-1}X^{r-1}$, where $\{a_i\}_{0\le i\le r-1}$ are the coefficients in $\mathbb{F}_p$. For elements β in $\mathrm{GR}(p^s, r)$ of the patter: $b_0 + b_1X + \cdots + b_{r-1}X^{r-1}$, where $b_i \in \{p \cdot q + a_i \mid q \in [0, p^{s-1} - 1]\}$, $\pi(\beta) = y$ and $\beta \in \mathrm{GR}^{\times}(p^s, r)$. When T_i is uniformly sampled from $\mathrm{GR}^{\times}(p^s, r)$, for any $\alpha \in \mathbb{F}_{p^r}^{\times}$, the probability that $\pi(T_i) = \alpha$ is:

$$\Pr(\pi(T_i) = \alpha) = \frac{|\{x \in \mathrm{GR}^{\times} \mid \pi(x) = \alpha\}|}{|\mathrm{GR}^{\times}|} = \frac{|\beta|}{|\mathrm{GR}^{\times}|} = \frac{p^{r(s-1)}}{p^{r(s-1)}(p^r - 1)} = \frac{1}{|\mathbb{F}_{p^r}^{\times}|}.$$

This confirms that $\pi(T_i)$ is uniformly distributed over $\mathbb{F}_{p^r}^{\times}$. Since we assume the folding elements T'_i for random codes over $\mathbb{F}_{p^r}$ are also uniformly sampled from $\mathbb{F}_{p^r}^{\times}$, the distribution of $\pi(T_i)$ matches the distribution of T'_i. Thus, C'_d is indeed a (c, k_0, d)-foldable random linear code over $\mathbb{F}_{p^r}$ by definition 9 in [18]. And $\mathbf{G'_i} = \pi(\mathbf{G_i})$, so $\mathbf{m'G'_d} = \mathbf{mG_d} \mod p$. We can now apply Theorem 2 from [18] to bound the zero entries in C'_d, which is exactly the upper bound of $\mathsf{num0}(e_p)$ and consequently we have the lower bound of Δ_{C_d}. We show concrete code distances in table 1, and $t_0 = 1$ by definition 10.

Table 1: Practical performance on the relative minimum distance of random foldable codes over Galois rings $\mathrm{GR}(p^s, r)$. The security parameter λ is 128.

k_0	k_d	c	$\lvert p^r \rvert$	Δ_{C_d}
1	2^8	16	2^{32}	0.4176
1	2^{10}	16	2^{64}	0.6728
1	2^{10}	8	2^{128}	0.6760
1	2^{25}	16	2^{128}	0.7144
1	2^{25}	8	2^{256}	0.7288

[3] This can easily be verified by Definition 10, as $\mathbf{G'_0}$ can be viewed as a generator matrix for a Reed-Solomon code over a finite field.

4 Polylogarithmic Polynomial Commitment

In this section, we generalize the Basefold IOPP for foldable codes over Galois rings $\mathrm{GR}(p^s, r)$ and combined with a sum-check protocol over Galois rings $\mathrm{GR}(p^s, r)$, construct a multilinear polynomial commitment scheme with logarithmic verification. Importantly, this construction applies to any foldable code over Galois rings, not just the random foldable codes introduced in the previous section. For the random foldable codes we achieve $\mathcal{O}(n \log n)$ encoding time.

4.1 BaseFold IOPPs for Foldable Codes over Galois Rings

For some integer $d \geq 0$, a foldable linear code C_d over $\mathrm{GR}(p^s, r)$ is specified by a sequence of generator matrices $(\mathbf{G_0}, \mathbf{G_1}, \ldots, \mathbf{G_d})$, where for each $i \in [1, d]$, $\mathbf{G_i} \in \mathrm{GR}(p^s, r)^{k_i \times n_i}$ has the following structure:

$$\mathbf{G_i} = \begin{bmatrix} \mathbf{G_{i-1}} & \mathbf{G_{i-1}} \\ \mathbf{G_{i-1}} \cdot T_{i-1} & \mathbf{G_{i-1}} \cdot T'_{i-1} \end{bmatrix}$$

where $T_{i-1}, T'_{i-1} \in \mathrm{GR}(p^s, r)^{n_{i-1} \times n_{i-1}}$ are diagonal matrices. For each index $j \in [n_{i-1}]$, we require the diagonal difference $\mathsf{diag}(T_{i-1})[j] - \mathsf{diag}(T'_{i-1})[j]$ is invertible in $\mathrm{GR}(p^s, r)^{n_{i-1}}$. This requirement stems from the fact that interpolation is not directly applicable over a ring. The Basefold folding process relies on linear interpolation, where deriving a third point on a line from two known points on the same line. By definition 10, $T'_{i-1} = -T_{i-1}$ and T_{i-1} sampled from $\mathrm{GR}^{\times}(p^s, r)$, we have $\mathsf{diag}(T_{i-1})[j] - \mathsf{diag}(T'_{i-1})[j] = 2\mathsf{diag}(T_{i-1})[j]$, which is invertible in $\mathrm{GR}(p^s, r)$ if $p \neq 2$. This is analogous to the requirement in Basefold [18] that $2\mathsf{diag}(T_{i-1})[j]$ is invertible in a field of characteristic not 2. The goal of IOPP is to verify that a given vector $\pi_d \in \mathrm{GR}(p^s, r)^{n_d}$ is close to a codeword in C_d. We present the protocol in two phases: commit and query in Figure 1

Completeness guarantees that if π_d is a codeword in C_d, and the prover follows the commit protocol honestly, then the verifier always accepts in the query phase. This property stems from the foldability and linearity of the code.

Soundness guarantees that if π_d is significantly distant from any valid codeword in the code C_d, then with overwhelming probability after the IOPP.Commit phase, the initial oracle π_0 will also be demonstrably distant from any codeword in C_0. Consequently, during the IOPP.Query phase, the verifier will output accept with only negligible probability.

The central challenge in establishing soundness lies in ensuring a sufficient distance preservation gap throughout the IOPP.Commit phase. This requires a quantifiable proximity gap for the underlying random linear codes over Galois rings, a concept frequently utilized in the proximity error analysis of IOPP protocols[5,4]. Existing literature, however, lacks a proven distance preservation gap specifically for linear codes defined over Galois rings. To address this, we present the first proximity gap analysis (Theorem 1), extending Theorem 4.4 from [5]. This theorem serves as the foundation for guaranteeing the soundness of our proposed IOPP protocol. Free module over rings play a role like linear space over fields.

IOPP.Commit Phase

Input Oracles: $\pi_d \in \mathrm{GR}(p^s, r)^{n_d}$
Output Oracles: $(\pi_{d-1}, \ldots, \pi_0)$.

From $i = d-1$ down to 0:

1. Verifier: sample $\alpha_i \leftarrow \mathrm{GR}(p^s, r)$, send α_i to prover.

2. For each $j \in [1, n_{i-1}]$:
Prover:
construct a polynomial $f(X)$ of degree ≤ 1 over $\mathrm{GR}(p^s, r)$ as:

$$f(X) = \pi_{i+1}[j] + \frac{\pi_{i+1}[j] - \pi_{i+1}[j+n_i]}{\mathsf{diag}(T_i)[j] - \mathsf{diag}(T_i')[j]}(X - \mathsf{diag}(T_i)[j]).$$

Set $\pi_i[j] = f(\alpha_i)$.

3. Prover: send oracle $\pi_i \in \mathrm{GR}(p^s, r)^{n_i}$ to verifier.

IOPP.Query Phase

Oracles: $(\pi_d, \ldots, \pi_1, \pi_0)$.

1. Verifier: sample query index $\mu \leftarrow [1, n_{d-1}]$.

2. From $i = d-1$ down to 0,
Verifier:

Query oracles $\pi_{i+1}[\mu]$ and $\pi_{i+1}[\mu + n_i]$ and compute $p(X)$ as:

$$p(X) = \pi_{i+1}[\mu] + \frac{\pi_{i+1}[\mu] - \pi_{i+1}[\mu+n_i]}{\mathsf{diag}(T_i)[\mu] - \mathsf{diag}(T_i')[\mu]}(X - \mathsf{diag}(T_i)[\mu]).$$

Check if $p(\alpha_i) = \pi_i[\mu]$. If not, Reject;

If $i > 0$ and $\mu \geq n_{i-1}$, update $\mu = \mu - n_{i-1}$.

3. Verifier: if π_0 is a valid codeword w.r.t. G_0 over $\mathrm{GR}(p^s, r)$, Accept; else Reject.

Fig. 1: IOPP for foldable codes over $\mathsf{GR}(p^s, r)$

Theorem 1 (Linear proximity gap for linear codes over Galois Rings). *Let $V \subseteq GR(p^s, r)^n$ be a free module over the Galois ring $GR(p^s, r)$ with $\Delta_V = \zeta$. Let $u^* \in GR(p^s, r)^n$ and let $\gamma > 0$ satisfy $\delta < J_\gamma(J_\gamma(\zeta))$. For $u \in GR(p^s, r)^n$, define $A = A_{u,\gamma} = \{\alpha \in GR(p^s, r) \mid \Delta(u^* + \alpha u, V) < \delta - \gamma\}$. If $|A| > \frac{2p^{r(s-1)}}{\gamma^3}$, then there exist $v^*, v \in V$ such that $|\{i \in [n] \mid (u_i = v_i) \wedge (u^* = v^*)\}| \geq (1-\delta)n$.*

Theorem 2 (Soundness adapted from Theorem 3[18]). *Let C_d be a (c, k_0, d) -foldable linear code defined over Galois ring $GR(p^s, r)$. Let C_i denote the code generated by $\mathbf{G}_i$ for $0 \leq i \leq d$, with $\Delta_{C_{i+1}} < \Delta_{C_i}$. Define a constant $\gamma > 0$ and set $\delta := \min(\Delta^*(\pi_d, C_d), J_\gamma(J_\gamma(\Delta_{C_d})))$, where $\Delta^*(\pi_d, C_d)$ is the relative coset minimum distance (Definition 4). Then, with a probability of at least $1 - \frac{2d}{\gamma^3 p^r}$*

over the randomly sampled challenges $(\alpha_0, \ldots, \alpha_{d-1})$ *in the IOPP.commit phase, for any prover oracles* $\{\pi_{d-1}, \ldots, \pi_0\}$*, the probability that the verifier outputs "accept" in all* l *repetitions of the IOPP.query phase is upper-bounded by* $(1-\delta+\gamma d)^l$*.*

We present proofs for Theorem 1 and Theorem 2 in Appendix A.1 and A.2.

Remark 1. We remove the interactions in our contruction using the Fiat-Shamir Heuristic [9,3] in the random oracle model. The transformation only incurs a negligible soundness loss [3]. Specifically, vectors sent by the prover in the IOP can be substituted with Merkle commitments to those vectors. Correspondingly, each query from the verifier to a vector is answered by the prover by providing the requested value along with a Merkle tree authentication path.

4.2 Sumcheck Protocol over Galois rings

Let us briefly review sum-check protocol ([13], Chapter 4 from [16]). To efficiently verify equation $\sum_{\mathbf{b}\in\{0,1\}^l} f(\mathbf{b}) = H$ over all binary inputs, where $f : \mathbb{F}^l \to \mathbb{F}$ is a d-degree multivariate polynomial, the prover and verifier engage in an l-round protocol with soundness error $\frac{dl}{|\mathbb{F}|}$. In each round i, the prover sends a univariate polynomial f_i, and the verifier responds with a random challenge r_i, checking that $f_i(0) + f_i(1)$ equals a given value. In the final round, the verifier queries the prover for the evaluation of $f(r_1, \ldots, r_l)$ and outputs accept or reject. The sum-check protocol over Galois rings presents in Figure 2.

Sumcheck Protocol over Galois Ring $\mathbf{GR}(p^s, r)$

Input: $\sum_{b_1,b_2,\ldots,b_l\in\{0,1\}} f(b_1, b_2, \ldots, b_l) = H$.

For $i = 1$:

Prover: send univariate $f_1(x_1) = \sum_{b_2,\ldots,b_l\in\{0,1\}} f(x_1, b_2, \ldots, b_l)$ to verifier.
Verifier: check $H = f_1(0) + f_1(1)$, send random $r_1 \in \mathrm{GR}(p^s, r)$ to prover.

From $i = 2$ to $l - 1$:

Prover: send univariate $f_i(x_i) = \sum_{b_{i+1},\ldots,b_l\in\{0,1\}} f(r_1, \ldots, r_{i-1}, x_i, b_{i+1} \ldots, b_l)$ to verifier.
Verifier: check $f_{i-1}(r_{i-1}) = f_i(0) + f_i(1)$, send random $r_i \in \mathrm{GR}(p^s, r)$ to prover.

For $i = l$:

Prover: send univariate $f_l(x_l) = f(r_1, r_2, \ldots, r_{l-1}, x_l)$ to verifier.
Verifier: check $f_{l-1}(r_{l-1}) = f_l(0) + f_l(1)$, send random $r_l \in \mathrm{GR}(p^s, r)$ to prover.

Assume an oracle access to an evaluation $f(r_1, r_2, \ldots, r_l)$ of f. The verifier accepts if and only if the evaluation at the final round satisfies: $f_l(r_l) = f(r_1, r_2, \ldots, r_l)$. Note that the specific instantiation of this oracle is determined by the application of the sumcheck protocol.

Fig. 2: Sumcheck Protocol

Soundness of Sumcheck Protocol In the sumcheck protocol for multivariate polynomial $f : \mathrm{GR}^l \to \mathrm{GR}$ over $\mathrm{GR}(p^s, r)$, the verifier can sample challenges from $\mathrm{GR}(p^s, r)$. If the prover is dishonest, in at least one round the claimed polynomial is not correct, and by fact 2 the probability that the verifier accepts in a given round is at most $\frac{\deg(f)}{p^r}$. Since the total degree is at most d and there are l rounds, a union bound shows that the overall soundness error is $\frac{dl}{p^r}$.

4.3 Multilinear Polynomial Commitment

We present a multilinear polynomial commitment scheme over Galois rings with polylogarithmic proof size and verification time in Figure 3. The construction combines the Basefold IOPP (Figure 1) and sumcheck protocol (Figure 2), and the coefficients of f are ordered consistently in both protocols.

Completeness For an honest prover, denote f_i as the i-variate multilinear polynomial $f(X_1, \ldots, X_i, r_i, \ldots, r_{d-1})$ such that $\mathsf{Enc}_i(\mathbf{f}_i) = \pi_i$ in the evaluation protocol, where $\mathbf{f}_i \in \mathrm{GR}(p^s, r)^{2^i}$ as coefficients. For $i \in [0, d-1]$, by the definition 2, we can order the coefficients $\mathbf{f}_{i+1} = (\mathbf{f}_{i+1,l}, \mathbf{f}_{i+1,r})$ such that $\mathbf{f}_i = \mathbf{f}_{i+1,l} + r_i \cdot \mathbf{f}_{i+1,r}$, then $\mathbf{f}_0 = f(\mathbf{r})$ implies $\pi_0 \cdot \tilde{eq}_{\mathbf{z}}(\mathbf{r}) = \mathsf{Enc}_0(\mathbf{f}_0) \cdot \tilde{eq}_{\mathbf{z}}(\mathbf{r}) = \mathsf{Enc}_0(\mathbf{f}_0 \cdot \tilde{eq}_{\mathbf{z}}(r_0, \ldots, r_{d-1})) = \pi'_0 = \mathsf{Enc}_0(h_1(r_0))$ holds.

Binding Assume an adversary can open a commitment Ct to two distinct polynomials f_1 and f_2, with corresponding codewords π_1 and π_2. This implies the Merkle commit of π_1 and π_2 are the same. Consequently, either $\pi_1 = \pi_2$ or the adversary has found a hash collision. If $\pi_1 = \pi_2$, then given that $\Delta^*(\mathsf{Enc}_d(\mathbf{f}_1), \pi_1)$, $\Delta^*(\mathsf{Enc}_d(\mathbf{f}_2), \pi_1) < \frac{\Delta_{C_d}}{2}$, by triangle inequality and lemma 6 in [18], we have $\Delta(\mathsf{Enc}_d(\mathbf{f}_1), \mathsf{Enc}_d(\mathbf{f}_2)) \leq \Delta^*(\mathsf{Enc}_d(\mathbf{f}_1), \mathsf{Enc}_d(\mathbf{f}_2)) < \Delta_{C_d}$, therefore $f_1 = f_2$.

Theorem 3 (Soundness of Polynomial Commitment). *Let $\gamma, \delta \in (0, 1)$ such that $3\delta - \gamma d < \Delta_{C_d}$ and $\frac{2d}{\gamma^3 p^r} + (1 - \delta + \gamma d)^l < 2^{-\lambda}$, where λ is the security parameter. Let C_i denote the code generated by $\mathbf{G}_i$ with $\Delta_{C_{i+1}} < \Delta_{C_i}$ for $0 \leq i \leq d$. In the evaluation protocol (Figure 3), for any malicious prover, the probability that verifier passes the checks in Step 3 is upper-bounded by $\frac{2d}{p^r}$.*

We present the proof for Theorem 3 and Extractablity in Appendix B.1 and B.2.

Prover time In commitment phase, the cost of encoding of multilinear polynomial f is dominated by $\mathcal{O}(n_d \log n_d) = \mathcal{O}(ck_d \log ck_d)$ multiplications over $\mathrm{GR}(p^s, r)$ and then hashes the ck_d elements of corresponding codeword. c is fixed constant defined in code C_0. In evaluation phase, the prover runs $\mathsf{IOPP.Commit}$ and sumcheck protocol. In sumcheck protocol, the prover computes polynomial $h_i(X)$, let T_h denote the cost of an evaluation of $f(\mathbf{b}, X, \mathbf{r}) \cdot \tilde{eq}_{\mathbf{z}}(\mathbf{b}, X, \mathbf{r})$ at point ($\mathbf{b}$,X,$\mathbf{r}$). Then the cost of computing $h_i(X)$ is $2^{i-1}T_h$, the totol cost of prover to compute $\{h_i\}_{1 \leq i \leq d}$ is $(2^d - 1)T_h$. T_h depends on the sparsity of specific f, since f is multilinear(the degree for each variable is 1 or 0), so if we precompute $\prod_{i=1}^{d} r_i$, then computing f and $\tilde{eq}_z$ only consists of constant additions over $\mathrm{GR}(p^s, r)$ and

Multilinear Polynomial Commitment over Galois Ring $\mathbf{GR}(p^s, r)$

Prover inputs: $f \in \mathrm{GR}(p^s, r)[X_1, X_2, \ldots, X_d]$ with coefficients $\mathbf{f} \in \mathrm{GR}(p^s, r)^{2^d}$.
Code Parameters: $\mathbf{G_0}$ and diagnal matrices $T_0, T_1, \ldots, T_{d-1}$ and $T'_0, T'_1, \ldots, T'_{d-1}$.
$\tilde{eq}_{b_1,\ldots,b_d}(X_1, \ldots, X_d) = \prod_{i=1}^{d}(b_i x_i + (1-b_i)(1-x_i))$ and $\tilde{eq}_{b_1,\ldots,b_d}(X_1, \ldots, X_d) = \tilde{eq}_{X_1,\ldots,X_d}(b_1, \ldots, b_d)$.

Commit Phase:

Prover: compute the oracle $\pi_f = \mathrm{Enc}_d(\mathbf{f})$ and send the oracle π_f as the commitment Ct to verifier.
(The derived commitment in the random oracle model is the root of merkle tree.)

Opening Phase:
For opening polynomial f given commitment Ct:

Prover: send polynomial f and a word π_f to verifier.

Verifier: perform the following checks:

- Check if the Merkle commitment of π_f equals Ct;
- Check if $\Delta(\pi_f, \mathsf{Enc}_d(f))$ is less than $\frac{\Delta_{C_d}}{2}$.

Evaluation Phase:
For opening polynomial f at point $\mathbf{z} \in \mathrm{GR}(p^s, r)^d$ with claimed value $y \in \mathrm{GR}(p^s, r)$ given commitment Ct:

1. Prover: send $h_d(X) = \sum_{\mathbf{b} \in \{0,1\}^{d-1}} f(\mathbf{b}, X) \cdot \tilde{eq}_{\mathbf{z}}(\mathbf{b}, X)$.

2. For $i = d-1$ down to 0:

Verifier: send random $r_i \in \mathrm{GR}(p^s, r)$ to prover.

Prover: perform as follows:

- for each $j \in [1, n_i]$, set $g_j(X)$ as:

$$g_j(X) = \pi_{i+1}[j] + \frac{\pi_{i+1}[j] - \pi_{i+1}[j+n_i]}{\mathsf{diag}(T_i)[j] - \mathsf{diag}(T'_i)[j]}(X - \mathsf{diag}(T_i)[j])$$

  and set $\pi_i[j] = g_j(r_i)$, send oracle $\pi_i \in \mathrm{GR}(p^s, r)^{n_i}$.
- if $i > 0$, send h_i as:

$$h_i(X) = \sum_{\mathbf{b} \in \{0,1\}^{i-1}} f(\mathbf{b}, X, r_i, \ldots, r_{d-1}) \cdot \tilde{eq}_{\mathbf{z}}(\mathbf{b}, X, r_i, \ldots, r_{d-1})$$

3. Verifier: perform the following checks:

- Check if $\mathsf{IOPP.Query}^{(\pi_d, \ldots, \pi_1, \pi_0)}$ outputs accept.
- Check if $h_d(0) + h_d(1) = y$ and for every $i \in [1, d-1]$, $h_i(0) + h_i(1) = h_{i+1}(r_i)$
- Check if $\mathsf{Enc}_0(h_1(r_0))$ equals π'_0, where $\pi'_0 = \pi_0 \cdot \tilde{eq}_{\mathbf{z}}(r_0, \ldots, r_{d-1}) = (\pi_{0,1} \cdot \tilde{eq}_{\mathbf{z}}(\mathbf{r}), \pi_{0,2} \cdot \tilde{eq}_{\mathbf{z}}(\mathbf{r}), \ldots, \pi_{0,c} \cdot \tilde{eq}_{\mathbf{z}}(\mathbf{r}))$.

Fig. 3: Multilinear Polynomial Commitment Scheme

compute $f \cdot \tilde{eq}_z$(two degree-1 polynomials) only envolves 4 multiplications and 1 addition over $\mathrm{GR}(p^s, r)$. Thus $(2^d - 1)T_h \approx 4(2^d - 1) \approx 4k_d$ multiplications. In $\mathsf{IOPP.Commit}$, the prover computes $\{g_j(r_i)\}_{1 \le j \le n_i}$ and obtain π_i then computes each merkle roots of π_i. These evolves $\mathcal{O}(ck_i)$ ring operations and hashes. So over all d rounds, the prover complexity is $\mathcal{O}(ck_d) + \mathcal{O}(4k_d) \approx \mathcal{O}(k_d)$.

Proof size The proof size consists of $\{\mathsf{Merkle.Commit}(\pi_i)\}$, $\{h_i(X)\}$ and $\{r_i\}$, the merkle roots are outputs of hash functions where one output is hundreds of bits, and $\{h_i(X)\}$ consists at most 3 ring elements since $h_i(X)$ is degree-2 polynomials. Additionally in the $\mathsf{IOPP.Query}$ phase, μ and a merkle tree authentication path of each oracle π_i, the path for the opening of π_i consists of $\log n_i$ hash outputs. And there are $\log n_d$ oracles to be opened, thus resulting in $\mathcal{O}\big(\log^2 n_d\big) = \mathcal{O}\big(\log^2 ck_d\big) \approx \mathcal{O}\big(\log^2 k_d\big)$ complexity in proof size.

Verifier time Verifier needs to perform $\mathcal{O}(d)$ ring operations for sumcheck protocol and $\mathsf{IOPP.Query}$ phase, verifier needs to check the merkle tree authentication paths which for one path verificiton needs at most $\mathcal{O}(d)$ hash computations, thus for l runs of query, the verifier complexity if $\mathcal{O}\big(l \log^2 n_d\big) \approx \mathcal{O}\big(\log^2 k_d\big)$.

4.4 Multipoint Opening Protocol

Additionally, we give a method to further accelerate the efficiency of our polynomial commitment scheme when there are many polynomials and they are queried to evaluate at multiple points. For polynomials $\{f_i\}_{1 \le i \le N}$ and points $\{\mathbf{z}_j\}_{1 \le j \le M}$, we show the multipoint protocol in figure 4, which is designed for opening multiple polynomials at the same set of points.

In the evaluation phase in figure 4, when prover runs $\mathsf{IOPP.Commit}$ for the polynomial ϕ, it uses $\pi_{\phi,d}$ to compute $\pi_{\phi,d-1}$, so in the $\mathsf{IOPP.Query}$ the verifier has to query $\pi_{\phi,d}$, but verifier only has oracle accesses to $\{\pi_{f_i,d}\}_{1 \le i \le N}$. So we slightly change the query phase, on input oracles $\{\pi_{f_i,d}\}_{1 \le i \le N}, \{\pi_{\phi,i}\}_{d-1 \ge i \ge 0}$:

- Verifier sample query index $\mu \in [n_{d-1}]$.
- From $i = d - 1$ down to 0:
  If $i = d-1$, query oracles $\{\pi_{f_i,d}\}_{1 \le i \le N}$, compute $\pi_{\phi,d}[\mu] = \sum_{i=1}^{N} \alpha^{i-1} \pi_{f_i,d}[\mu]$ and $\pi_{\phi,d}[\mu + n_{d-1}] = \sum_{i=1}^{N} \alpha^{i-1} \pi_{f_i,d}[\mu + n_{d-1}]$; Else if $0 \le i \le d-2$, query $\pi_{\phi,i+1}$. Then compute $p(X)$ and check if $p(r_i) = \pi_{\phi,i}[\mu]$, update $\mu = \mu - n_{i-1}$.
- if $\pi_{\phi,0}$ is a valid codeword output accept; else reject

Efficiency Analysis The commitment cost is simply N times the cost of committing to a polynomial of length k_d. In the evaluation phase of the sum-check protocol, the prover needs to additionally compute M instances of $\tilde{eq}_{\mathbf{z}_j}$ and their linear combination per round. The proof size in the $\mathsf{IOPP.Query}$ phase only increases by N openings for $\{\pi_{f_i,d}\}_{1 \le i \le N}$. The verifier's cost only increases by the hash computation during the openings of $\{\pi_{f_i,d}\}_{1 \le i \le N}$ and the computation of $\sum_{j=1}^{M} \beta^{j-1} \tilde{eq}_{\mathbf{z}_j}(r_0, \ldots, r_{d-1})$.

Multipoint Opening of Polynomial Commitments

Prover inputs: For N multilinear polynomials $f_1, \ldots, f_N \in \mathrm{GR}(p^s, r)[X_1, X_2, \ldots, X_d]$ with coefficients $\mathbf{f}_1, \ldots, \mathbf{f}_N \in \mathrm{GR}(p^s, r)^{2^d}$.
Code Parameters: $\mathbf{G_0}$ and diagnal matrices $T_0, T_1, \ldots, T_{d-1}$ and $T'_0, T'_1, \ldots, T'_{d-1}$.

Commit Phase:

Prover: For $1 \le i \le N$, compute the oracle $\pi_{f_i,d} = \mathrm{Enc}_d(\mathbf{f}_i)$ and send the oracles $\{\pi_{f_i,d}\}_{1\le i\le N}$ as the commitment $\{\mathsf{Ct}_i\}_{1\le i\le N}$ to verifier

Verifier: send random $\alpha, \beta \in \mathrm{GR}(p^s, r)$ to prover.

Define: $\phi = \sum_{i=1}^{N} \alpha^{i-1} f_i$, $\pi_{\phi,d} = \sum_{i=1}^{N} \alpha^{i-1} \pi_{f_i,d} = \mathsf{Enc}_d(\phi)$(Linear Code).

Evaluation Phase:
For opening polynomials $\{f_i\}_{1\le i\le N}$ at points $\{\mathbf{z}_j\}_{1\le j\le M}$ for each $\mathbf{z}_j \in \mathrm{GR}(p^s, r)^d$ with claimed values $f_i(\mathbf{z}_j) = y_{ij} \in \mathrm{GR}(p^s, r)$ given commitments $\{\mathsf{Ct}_i\}_{1\le i\le N}$:

1.Prover: send $h_d(X) = \sum_{\mathbf{b}\in\{0,1\}^{d-1}} (\phi(\mathbf{b}, X) \cdot \sum_{j=1}^{M} \beta^{j-1} \cdot \tilde{eq}_{\mathbf{z}_j}(\mathbf{b}, X))$.

2.For $i = d-1$ down to 0:

Verifier: send random $r_i \in \mathrm{GR}(p^s, r)$ to prover.

Prover: perform as follows:

- for each $j \in [1, n_i]$, set $g_j(X)$ as:

$$g_j(X) = \pi_{\phi,i+1}[j] + \frac{\pi_{\phi,i+1}[j] - \pi_{\phi,i+1}[j+n_i]}{\mathsf{diag}(T_i)[j] - \mathsf{diag}(T'_i)[j]}(X - \mathsf{diag}(T_i)[j])$$

  and set $\pi_{\phi,i}[j] = g_j(r_i)$, send oracle $\pi_{\phi,i} \in \mathrm{GR}(p^s, r)^{n_i}$.
- if $i > 0$, send h_i as:

$$h_i(X) = \sum_{\mathbf{b}\in\{0,1\}^{i-1}} (\phi(\mathbf{b}, X, r_i, \ldots, r_{d-1}) \sum_{j=1}^{M} \beta^{j-1} \cdot \tilde{eq}_{\mathbf{z}_j}(\mathbf{b}, X, r_i, \ldots, r_{d-1}))$$

3.Verifier: perform the following checks:

- Check if $\mathsf{IOPP.Query}^{(\{\pi_{f_i,d}\}_{1\le i\le N}, \{\pi_{\phi,i}\}_{d-1\ge i\ge 0})}$ outputs accept.
- Check if $h_d(0) + h_d(1) = \sum_{j=1}^{M} \beta^{j-1} \sum_{i=1}^{N} \alpha^{i-1} y_{ij}$ and for every $i \in [1, d-1]$, $h_i(0) + h_i(1) = h_{i+1}(r_i)$
- Check if $\mathsf{Enc}_0(h_1(r_0))$ equals $\pi'_{\phi,0}$, where $\pi'_{\phi,0} = \pi_{\phi,0} \cdot \sum_{j=1}^{M} \beta^{j-1} \tilde{eq}_{\mathbf{z}_j}(\mathbf{r}) = (\pi_{\phi,0,1} \cdot \sum_{j=1}^{M} \beta^{j-1} \tilde{eq}_{\mathbf{z}_j}(\mathbf{r}), \ldots, \pi_{\phi,0,c} \cdot \sum_{j=1}^{M} \beta^{j-1} \tilde{eq}_{\mathbf{z}_j}(\mathbf{r}))$.

Fig. 4: Multipoint opening Protocol

Soundness of Multipoint Opening Protocol Conditioned on the soundness of the sumcheck protocol holding within the evaluation protocol, and then assuming there exists $1 \le u \le N$ and $1 \le v \le M$ such that $f_u(\mathbf{z}_v) \ne y_{uv}$, the probability that the verifier passes the check is at most a negligible probability

plus the following probability:

$$
\begin{aligned}
&\Pr[h_d(0)+h_d(1)=\sum_{j=1}^{M}\beta^{j-1}\sum_{i=1}^{N}\alpha^{i-1}y_{ij}]\\
&\le \Pr[\sum_{\mathbf{b}\in\{0,1\}^d}(\phi(\mathbf{b})\cdot\sum_{j=1}^{M}\beta^{j-1}\cdot \tilde{eq}_{\mathbf{z}_j}(\mathbf{b}))=\sum_{j=1}^{M}\beta^{j-1}\sum_{i=1}^{N}\alpha^{i-1}y_{ij}]\\
&=\Pr[\sum_{j=1}^{M}\beta^{j-1}\sum_{\mathbf{b}\in\{0,1\}^d}(\phi(\mathbf{b})\cdot \tilde{eq}_{\mathbf{z}_j}(\mathbf{b}))=\sum_{j=1}^{M}\beta^{j-1}\sum_{i=1}^{N}\alpha^{i-1}y_{ij}]\\
&=\Pr[\sum_{j=1}^{M}\beta^{j-1}\sum_{\mathbf{b}\in\{0,1\}^d}(\sum_{i=1}^{N}\alpha^{i-1}f_i(\mathbf{b})\cdot \tilde{eq}_{\mathbf{z}_j}(\mathbf{b}))=\sum_{j=1}^{M}\beta^{j-1}\sum_{i=1}^{N}\alpha^{i-1}y_{ij}]\\
&=\Pr[\sum_{j=1}^{M}\beta^{j-1}\sum_{i=1}^{N}\alpha^{i-1}(\sum_{\mathbf{b}\in\{0,1\}^d}f_i(\mathbf{b})\cdot \tilde{eq}_{\mathbf{z}_j}(\mathbf{b})-y_{ij})=0]
\end{aligned}
$$

Let polynomial $A(x)=\sum_{i=1}^{N}x^{i-1}(\sum_{\mathbf{b}\in\{0,1\}^d}f_i(\mathbf{b})\cdot\tilde{eq}_{\mathbf{z}_v}(\mathbf{b})-y_{iv})$ over $\mathrm{GR}(p^s,r)$, we have $\sum_{\mathbf{b}\in\{0,1\}^d}f_u(\mathbf{b})\cdot\tilde{eq}_{\mathbf{z}_v}(\mathbf{b})-y_{uv}\neq 0$, so $A(x)$ is not a zero polynomial, by fact 2 we have $\Pr_{x\xleftarrow{\$}\mathrm{GR}(p^s,r)}[A(x)=0]\le\frac{N-1}{p^r}$. Let polynomial $B(x)=\sum_{j=1}^{M}x^{j-1}(\sum_{i=1}^{N}\alpha^{i-1}(\sum_{\mathbf{b}\in\{0,1\}^d}f_i(\mathbf{b})\cdot\tilde{eq}_{\mathbf{z}_j}(\mathbf{b})-y_{ij}))$.

Thus we have:

$$
\begin{aligned}
&\Pr[\sum_{j=1}^{M}\beta^{j-1}\sum_{i=1}^{N}\alpha^{i-1}(\sum_{\mathbf{b}\in\{0,1\}^d}f_i(\mathbf{b})\cdot\tilde{eq}_{\mathbf{z}_j}(\mathbf{b})-y_{ij})=0]=\Pr_{x\xleftarrow{\$}\mathrm{GR}(p^s,r)}[B(x)=0]\\
&=\Pr_{x,y\xleftarrow{\$}\mathrm{GR}(p^s,r)}[B(x)=0\mid A(y)=0]+\Pr_{x,y\xleftarrow{\$}\mathrm{GR}(p^s,r)}[B(x)=0\mid A(y)\neq 0]\\
&\le\frac{N-1}{p^r}+(1-\frac{N-1}{p^r})\cdot\frac{M-1}{p^r}\le\frac{M+N-2}{p^r}
\end{aligned}
$$

Thus if p^r is large enough such that $\frac{M+N-2}{p^r}<2^{-\lambda}$, a malicious prover can only success with negligible probability.

Remark 2 (Sample Challenges in Extension Rings). The soundness error of the aforementioned PCS, IOPP depend on p^r. Therefore, if p^r is not sufficiently large, resulting in a soundness error no smaller than $2^{-\lambda}$, we can sample challenges from a sufficiently large extension ring $\mathrm{GR}(p^s,rk)$ obtained by extending $\mathrm{GR}(p^s,r)$. This is a common practice in many SNARKs based on small fields. In this case, for an encoding $\mathrm{Enc}(\mathbf{m})$, we can treat $\mathbf{m}\in\mathrm{GR}(p^s,r)$ as a constant term in the extension ring $\mathrm{GR}(p^s,rk)$. Consequently, for a message $\mathbf{m}\in(\mathrm{GR}(p^s,r))^{k_d}$, the encoding of $\mathbf{m}$ over the small ring $\mathrm{GR}(p^s,r)$ is exactly the encoding of $\mathbf{m}$ over the extension ring $\mathrm{GR}(p^s,rk)$. Thus, the prover's computational cost in the commitment phase remains based on the small ring; we treat the commitment

as being over the large ring during the opening phase and then select random challenges in the extension ring to enhance soundness.

5 Conclusion

Combining our PCS with IOPs yields efficient zkSNARKs for polynomial ring arithmetic (Theorem 1 from [8]), significantly impacting real-world applications like verifiable fully homomorphic encryption (VFHE), enabling auditable privacy-preserving cloud computing, secure multi-party neural network inference, and regulatory-compliant financial derivatives calculations.

We also compare our work with existing Galois ring polynomial commitment schemes in Table 2, highlighting multi-opening efficiency gains through our batched protocol over independent evaluations.

Table 2: Comparisons of different polynomial commitment schemes over Galois rings. The first two rows compares a multilinear polynomial with n coefficients and evaluate on one point. The next two rows illustrate multipoint openings on N polynomials and M points. The last row show the complexity of our multipoint opening protocol.

Scheme	Commit Phase	Evaluation Phase		
	Commit time	Prover time	Proof size	Verifier time
$\mathsf{Brakedown}_{\mathrm{GR}}$[17,12]	$\mathcal{O}(n)$	$\mathcal{O}(n)$	$\mathcal{O}(\sqrt{n})$	$\mathcal{O}(\sqrt{n})$
$\mathsf{Basefold}_{\mathrm{GR}}$(**This Work**)	$\mathcal{O}(n \log n)$	$\mathcal{O}(n)$	$\mathcal{O}\big(\log^2 n\big)$	$\mathcal{O}\big(\log^2 n\big)$
This Work(Unbatched)	$\mathcal{O}(Nn \log n)$	$\mathcal{O}(NMn)$	$\mathcal{O}\big(NM \log^2 n\big)$	$\mathcal{O}\big(NM \log^2 n\big)$
This Work(Batched)	$\mathcal{O}(Nn \log n)$	$\mathcal{O}(n)$ $+\mathcal{O}(M \log n)$	$\mathcal{O}\big(\log^2 n\big)$ $+\mathcal{O}(N \log n)$	$\mathcal{O}\big(\log^2 n\big)$ $+\mathcal{O}(M)+$ $\mathcal{O}(N \log n)$

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A Soundness of IOPP for Foldable Codes

A.1 Proof of Theorem 1

Proof. For each $\alpha \in A$, there exists $v^\alpha \in V$ such that $\Delta(u^* + \alpha u, v^\alpha) < \delta - \gamma$. This implies that $\Delta(u^*, v^\alpha - \alpha u) < \delta - \gamma$. We construct a graph with vertex set A; two vertices α and α' are adjacent if $\Delta(v^\alpha - \alpha u, v^{\alpha'} - \alpha' u) < J_\gamma^{-1}(\delta)$.

By the Johnson bound from Theorem 2.2 [5], this graph cannot have an independent set of size $\frac{1}{\gamma}$. Applying Turán's theorem, we find a vertex α_0 with degree at least $\gamma|A| - 1$. Hence, there exists a subset $B \subseteq A$ with $|B| \geq \gamma|A| - 1$ such that for all $\alpha \in B$, $\Delta(v^{\alpha_0} - \alpha_0 u, v^\alpha) \leq J_\gamma^{-1}(\delta)$.

Define $C = C_{\alpha_0,N} = \{\alpha \in B \mid \alpha - \alpha_0 \notin (p)\}$. (p) denotes the maximal ideal consisting of non-invertible elements. For all $\alpha \in C$, we have $\Delta(u, \frac{1}{\alpha-\alpha_0}(v^\alpha - v^{\alpha_0})) \leq J_\gamma^{-1}(\delta)$. Since V has distance ζ and $J_\gamma(\zeta) > J_\gamma^{-1}(\delta)$, we apply the Johnson bound again, there can be at most $\frac{1}{\gamma}$ distinct vectors $v \in V$ such that $\Delta(u, v) < J_\gamma^{-1}(\delta)$. By the pigeonhole principle, there exists a vector $v \in V$ and a subset $D \subseteq C$ with $|D| \geq \gamma|C|$ such that for all $\alpha \in D$, we have $v = \frac{1}{\alpha-\alpha_0}(v^\alpha - v^{\alpha_0})$. Thus, for all $\alpha \in D$, we have $v^\alpha = (v^{\alpha_0} - \alpha_0 v) + \alpha v$.

Denote $v^\alpha = v^* + \alpha v$, where $v^*, v \in V$. Substituting back into the definition of v^α, we obtain for all $\alpha \in D$: $\Delta(u^*, v^* + \alpha(v - u)) < \delta - \gamma$. Rewriting gives: $\Delta(u^* - v^*, \alpha(v - u)) < \delta - \gamma$. For any coordinate $i \in [n]$ where $u_i \neq v_i$ or $u_i^* \neq v_i^*$, there can be at most $p^{r(s-1)}$ values of $\alpha \in \mathrm{GR}(p^s, r)$ satisfying $u_i^* - v_i^* = \alpha(v_i - u_i)$ (fact 1). Consequently, there must exist an $\alpha \in D$ such that:

$$\Delta(u^* - v^*, \alpha(v - u)) \geq 1 - \frac{|\{i \in [n] \mid (u_i = v_i) \wedge (u_i^* = v_i^*)\}|}{n} - \frac{p^{r(s-1)}}{|D|}.$$

Combining this with our upper bound on $\Delta(u^* - v^*, \alpha(v - u))$, we conclude:

$$\frac{|\{i \in [n] \mid (u_i = v_i) \wedge (u_i^* = v_i^*)\}|}{n} \geq 1 - \delta + \gamma - \frac{p^{r(s-1)}}{|D|}.$$

From the inequality $|D| \geq \gamma|C| = \gamma(|B| - p^{r(s-1)}) \geq \gamma^2|A| - \gamma(p^{r(s-1)} + 1)$, and given that:

$$|A| \geq \frac{2p^{r(s-1)}}{\gamma^3} \geq \frac{p^{r(s-1)}}{\gamma^3} + \frac{p^{r(s-1)} + 1}{\gamma},$$

with the easily satisfied condition $\gamma \leq \sqrt{\frac{p^{r(s-1)}}{p^{r(s-1)}+1}}$, we deduce: $\frac{|D|}{p^{r(s-1)}} \geq \frac{1}{\gamma}$. This completes the proof. □

A.2 Proof of Theorem 2

Proof. To bound the probability that a malicious prover significantly reduces the Hamming distance to the codeword space in the IOPP.commit phase, we analyze a single interaction round (where the verifier sends a random challenge $\alpha_i \in \mathrm{GR}(p^s, r)$ and the prover responds with π_i) and then apply a union bound

over all d rounds. Specifically, we bound the probability that $\Delta^*(\pi_i, C_i)$ is significantly smaller than $\Delta^*(\pi_{i+1}, C_{i+1})$. Let $\mathsf{fold}_\alpha(\pi_{i+1}) = \mathbf{u} + \alpha\mathbf{u}^*$ be the result of the folding operation with challenge α, where $\mathbf{u}$ and $\mathbf{u}^*$ are the unique interpolated vectors such that $\pi_{i+1} = (\mathbf{u} + \mathsf{diag}(T_i) \circ \mathbf{u}^*, \mathbf{u} + \mathsf{diag}(T_i') \circ \mathbf{u}^*)$. Define the set $A = \{\alpha \in \mathrm{GR}(p^s, r) \mid \Delta^*(\mathbf{u} + \alpha\mathbf{u}^*, C_i) < \delta^{(i+1)} - \gamma\}$. Here, $\delta^{(i)} = \min(\Delta^*(\pi_i, C_i), J_\gamma(J_\gamma(\Delta_{C_d}))$.

We next argue the probability that for randomly chosen challenge α_i such that $\alpha_i \in A$ is bounded by $\frac{2p^{r(s-1)}}{\gamma^3}$.

If $|A| > \frac{2p^{r(s-1)}}{\gamma^3}$, we know $\delta^{(i+1)} \leq J_\gamma(J_\gamma(\Delta_{C_d}))$ and $\Delta_{C_i} \geq \Delta_{C_{i+1}}$ for all $i \in [0, d-1]$, thus $\delta^{(i+1)} \leq J_\gamma(J_\gamma(\Delta_{C_i})))$, by Theorem 1, there exists $\mathbf{v}$ and $\mathbf{v}^*$ in C_i and a subset $S \subseteq [1, n_i]$, $|S| \geq (1 - \delta^{i+1})n_i$ such that $\mathbf{u}[s] = \mathbf{v}[s]$ and $\mathbf{u}^*[s] = \mathbf{v}^*[s]$ for all $s \in S$. By linearity of the code, $c_w = (\mathbf{v} + \mathsf{diag}(T_{i-1}) \circ \mathbf{v}^*, \mathbf{v} + \mathsf{diag}(T_{i-1}') \circ \mathbf{v}^*)$ is a codeword in C_{i+1}. Therefore, c_w agrees with π_{i+1} at positions s and $s + n_i$ for $s \in S$, thus $\Delta^*(\pi_{i+1}, C_{i+1}) < \frac{2\delta^{i+1}n_i}{n_{i+1}} = \delta^{(i+1)}$, which contracts $\Delta^*(\pi_{i+1}, C_{i+1}) \geq \delta^{(i+1)}$. Then, the probability a randomly chosen α_i belongs to A is upper bounded by:

$$P(\alpha_{d-1} \in A) = \frac{|A|}{|\mathrm{GR}(p^s, r)|} \leq \frac{\frac{2p^{r(s-1)}}{\gamma^3}}{p^{rs}} = \frac{2}{\gamma^3 p^r}$$

This implies that in round i, after the verifier sends challenge α_i, the probability that a cheating prover can successfully send an oracle π_i whose Hamming distance to C_i is notably smaller than $\Delta^*(\pi_{i+1}, \Delta_{C_{i+1}})$ is at most $\frac{2}{\gamma^3 p^r}$.

By applying the union bound over all d rounds, the probability that the verifier is cheated in any round $i \in [0, d-1]$ is bounded by $\sum_{i=0}^{d-1} \frac{2}{\gamma^3 p^r} = \frac{2d}{\gamma^3 p^r}$. Thus, with probability at most $\frac{2d}{\gamma^3 p^r}$, for all rounds of folding, there exists one round that $\Delta(\mathsf{fold}_{\alpha_i}(\pi_{i+1}), C_i) < \delta^{i+1} - \gamma$. We can demonstrate that the rejection probability in the $\mathsf{IOPP.Query}$ phase is upper-bounded by the expression $P = \sum_{i=0}^{d-1} \Delta(\pi_i, \mathrm{fold}_{\alpha_i}(\pi_{i+1}))$. Claim 2 of [18] then implies that $P \geq \delta - \gamma d$. Thus considering l repetitions of the query phase, the probability that the verifier outputs accept in all l repetitions upper-bounded by $(1 - \delta + \gamma d)^l$. □

B Soundness of Polynomial Commitment

B.1 Proof of Theorem 3

Proof. By IOPP soundness(Theorem 2), with overwhelming probability we have $\Delta^*(\pi_d, C_d) < \delta$, by lemma 6 in [18] we have $\Delta(\pi_d, C_d) \leq \Delta^*(\pi_d, C_d) \leq \delta < \frac{1}{2}\Delta_{C_d}$, thus there exists a unique vector $\mathbf{f} \in \mathrm{GR}(p^s, r)^{k_d}$ such that $\Delta(\mathsf{Enc}_d(\mathbf{f}), \pi_d) < \frac{1}{2}\Delta_{C_d}$. For the unique multilinear polynomial f with coefficients $\mathbf{f}$, assume that $f(\mathbf{z}) \neq y$, we prove that for any malicious prover, the probability that verifier passes the check in step **3** of evaluation protocol is negligible.

First we address the probability of $A = \{\pi_0 \neq \mathsf{Enc}_0(f(\mathbf{r})) \wedge \pi_0 \in C_0\}$ is negligible, for the random challenges $\mathbf{r} = (r_1, .., r_{d-1})$ in the evaluation protocol. Let f_i be the i-variate multilinear polynomial $f(X_1, \ldots, X_i, r_i, \ldots, r_{d-1}) \in$

$\mathrm{GR}(p^s, r)[X_1, X_2, \ldots, X_i]$. Since $\Delta^*(\pi_d, C_d) \leq \delta$, if A holds, then there exists at least one round $k \in [0, d-1]$ such that $\Delta^*(\pi_k, \mathsf{Enc}_k(\mathbf{g}_k)) < \delta$ for $\mathbf{g}_k \neq \mathbf{f}_k$, $g_k \in \mathrm{GR}(p^s, r)[X_1, X_2, \ldots, X_k]$ and $\Delta^*(\pi_{k+1}, \mathsf{Enc}_{k+1}(\mathbf{f}_{k+1})) < \delta$. Then by definition of minimum relative distance, $\Delta(\mathsf{Enc}_k(\mathbf{g}_k), \mathsf{Enc}_k(\mathbf{f}_k)) \geq \Delta_{C_k}$, and the different positions between π_{k+1} and $\mathsf{Enc}_{k+1}(\mathbf{f}_{k+1})$ is at most $n_{k+1} \cdot \delta$, thus:

$$\begin{aligned}
&\Delta^*(\mathsf{fold}_{r_k}(\pi_{k+1}), \mathsf{Enc}_k(\mathbf{f}_k)) \\
&= \Delta^*(\mathsf{fold}_{r_k}(\pi_{k+1}), \mathsf{fold}_{r_k}(\mathsf{Enc}_{k+1}(\mathbf{f}_{k+1}))) \\
&< \frac{n_{k+1}\delta}{n_k} = 2\delta < \Delta_{C_d} < \Delta_{C_k}
\end{aligned}$$

Again by IOPP soundness, we have $\Delta^*(\mathsf{fold}_{r_k}(\pi_{k+1}), C_k) < \delta - \gamma$, and according to above inequality we have $\Delta^*(\mathsf{fold}_{r_k}(\pi_{k+1}), \mathsf{Enc}_k(\mathbf{f}_k)) < \Delta_{C_k}$, since Δ_{C_k} is the minimum relative distance between any codewords in C_k, thus $\mathsf{Enc}_k(\mathbf{f}_k)$ is the only codeword satisfy $\Delta^*(\mathsf{fold}_{r_k}(\pi_{k+1}), \mathsf{Enc}_k(\mathbf{f}_k)) < \delta - \gamma < \delta$. Then by triangle inequality and $3\delta - \gamma d < \Delta_{C_d}$, we have:

$$\begin{aligned}
&\Delta(\pi_k, \mathsf{fold}_{r_k}(\pi_{k+1})) \\
&\geq \Delta(\pi_k, \mathsf{Enc}_k(\mathbf{f}_k)) - \Delta(\mathsf{Enc}_k(\mathbf{f}_k), \mathsf{fold}_{r_k}(\pi_{k+1})) \\
&\geq \Delta(\mathsf{Enc}_k(\mathbf{g}_k), \mathsf{Enc}_k(\mathbf{f}_k)) - \Delta(\pi_k, \mathsf{Enc}_k(\mathbf{g}_k)) - \Delta(\mathsf{Enc}_k(\mathbf{f}_k), \mathsf{fold}_{r_k}(\pi_{k+1})) \\
&\geq \Delta_{C_k} - \delta - \delta \geq \Delta_{C_d} - 2\delta > \delta - \gamma d
\end{aligned}$$

Thus in the $\mathsf{IOPP.Query}$ phase, if A holds, the accepting probability of verifier over l repetition runs is at most $(1 - (\Delta_{C_d} - 2\delta))^l < (1 - (\delta - \gamma d))^l < 2^{-\lambda}$ (by $3\delta - \gamma d < \Delta_{C_d}$), which is negligible. Finally, under the condition that $\pi_0 = \mathsf{Enc}_0(f(\mathbf{r}))$, the validity of the equation $\mathsf{Enc}_0(h_1(r_0)) = \pi_0 \cdot \tilde{eq}_{\mathbf{z}}(r_0, \ldots, r_{d-1})$ implies the validity of the equation $h_1(r_0)) = f(\mathbf{r}) \cdot \tilde{eq}_{\mathbf{z}}(r_0, \ldots, r_{d-1}$. If $f(\mathbf{z}) = \sum_{\mathbf{b} \in \{0,1\}^d} f(\mathbf{b}) \cdot \tilde{eq}_{\mathbf{z}}(\mathbf{b}) \neq y$, by soundness of sumcheck protocol, for any malicious prover, the probability that verifier passes the check in step **3** is at most $\frac{2d}{p^r}$, where the individual degree of each variable of $f(X)$ and $\tilde{eq}_{\mathbf{z}}(X)$ is 1 and the univariate polynomial $h_i(X)$ has degree 2 of variable X. □

B.2 Extractability

Knowledge soundness involves extraction to obtain an invertible matrix. Gaussian elimination can then be applied to recover coefficients of the multilinear polynomial used in the commitment scheme. For example, given $g(x, y) = a_1 + a_2 x + a_3 y + a_4 xy$, recovering $\mathbf{g} = (a_1, a_2, a_3, a_4)$ requires four evaluations of $g(x, y)$. However the presence of zero divisors among the monomials of a multilinear polynomial over Galois rings can lead to a non-invertible matrix. We next analyze the probability of invertible matrices. For a d-variate multilinear polynomial f, the probability that randomly chosen points $r_1, r_2, \ldots, r_d$ in the evaluation protocol are invertible elements is at most $\xi = (1 - \frac{d}{p^r})$, as $\Pr[r_i \in \mathrm{GR}(p^s, r) \setminus \mathrm{GR}^\times(p^s, r)] = \frac{1}{p^r}$. Therefore, in polynomial time $\mathcal{O}(2^d \cdot \frac{1}{\xi})$, one can extract a matrix consisting of invertible elements, ensuring the matrix is invertible, and enabling successful recovery via Gaussian elimination.